

Top 10 Operating Systems for Ethical Hackers and Penetration Testers

A hacker is a highly skilled computer operator who uses bugs and exploits to break into computer systems and networks. An ethical hacker, on the other hand, identifies vulnerabilities in computer systems and networks, and plugs these holes. Here's a round-up of the ten best operating systems for ethical hackers and penetration testers.

Hacking has been part of computing for over five decades and is a very broad area, covering a range of tasks. The term *hacker* originated in 1960 at MIT. Hacking is regarded as a process of locating all possible back doors that exist in a computer system or network and, finally, intruding into it. In simple terms, it is regarded as an unauthorised activity to gain access to a computer system or network, to harm it or prevent its normal functions and to steal sensitive information available on computing systems.

On the other hand, hacking can also be a legal activity when it is used to discover back doors and weaknesses in computer systems in order to patch them. This is called ethical hacking.

In today's cyber world, lots of varied attacks occur every hour and are achieved by professional hackers, virus writers and crackers. The most basic tool required by every hacker is a hacking operating system. Usually, the operating systems specialising in hacking are based on the Linux kernel. These operating systems are regarded as advanced working systems, which include lots of ethical hacking and penetration testing tools.

In this article, the ten best operating systems used by ethical hackers and penetration testers are featured. All of them are free, open source, are based on the Linux kernel and are bundled with many hacking tools.

1. Kali Linux

Kali Linux is based on the Debian-Linux distribution, and is especially designed for digital forensics and

penetration testing. It is maintained and updated on a regular basis by Offensive Security Ltd, under the leadership of Mati Aharoni, Devon Keams and Raphael Hertzog who are core developers. Kali comes pre-installed with more than 300 penetration testing programs and can be installed as a primary operating system on the hard disk, live CD/USB and can even run as a virtual machine using some virtualisation software.

Kali Linux supports both 32-bit and 64-bit images for use with x86 machines, and even supports various development boards like Raspberry Pi, BeagleBone, Odroid, CuBox, etc.

Tool categories

- **Information gathering:** These tools are used to gather information regarding DNS, IDS/IPS, network scanning, operating system, routing, SSL, SMB, VPN, VOPI, SNMP, email and VPN.
- **Vulnerability assessment:** These tools access the CISCO network and discover vulnerabilities in several database servers.
- **Web applications:** These tools are related to Web applications like content management systems, database exploitation, application proxies, Web vulnerability scanners, etc.
- **Password attacks:** These tools perform password attacks via brute force, dictionary attacks, etc.
- **Exploitation tools:** These exploit vulnerabilities in target systems for networks, the Web and databases, and to perform social engineering attacks.